

**Brac University**  
**Department of Computer Science and Engineering**

CSE111: Programming Language II    Summer 2024    Quiz 3    Marks: 10

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| Name: | ID: | Section: |
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1. Trace the below code and generate outputs.

[10]

**Design Class:**

```
public class ComplexNumber {
    public static int real = 1, imag = 1;
    public int x, y;

    public ComplexNumber(int x, int y) {
        this.x = x;
        this.y = y;
        real += x;
        imag += y;
    }

    public void add(int n) {
        this.x = real + n;
        this.y = this.y + this.x + this.multiply();
        System.out.println(this.x + " + " + this.y);
    }

    public int multiply() {
        this.y = imag * this.y + this.x;
        System.out.println(this.x + " + " + this.y);
        real *= this.x;
        return this.y;
    }

    public void combine(ComplexNumber c) {
        c.y = this.imag * c.y + this.y;
        System.out.println(c.x + " + " + c.y);
    }
}
```

**Tester Class:**

```
public class TracingTest {
    public static void main(String[] args) {
        ComplexNumber c1 = new ComplexNumber(2, 3);
        c1.add(4);
        ComplexNumber c2 = new ComplexNumber(1, 2);
        c2.add(3);
        c1.combine(c2);
    }
}
```

| Outputs |
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